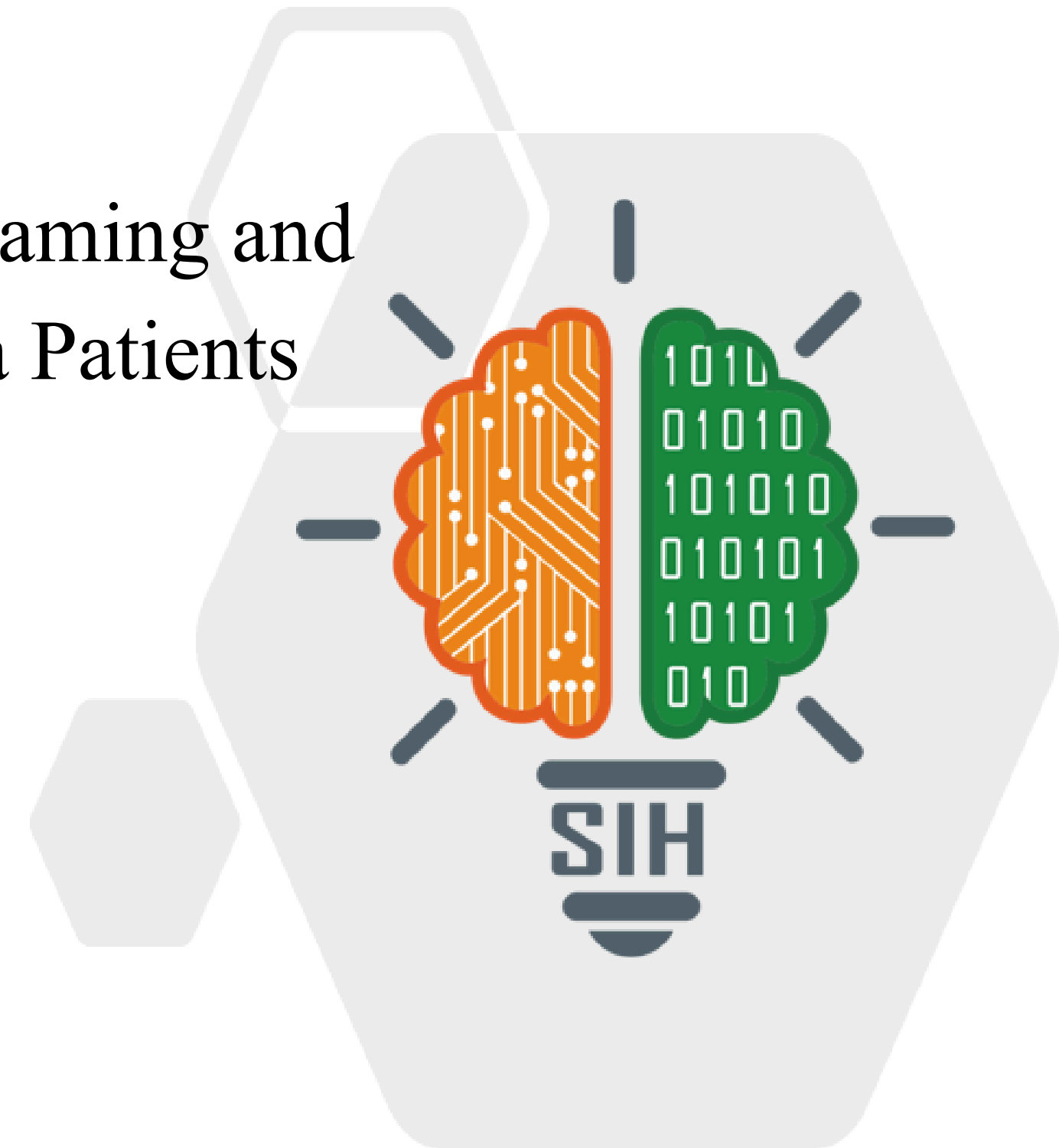


Team Name - Code Catalyst

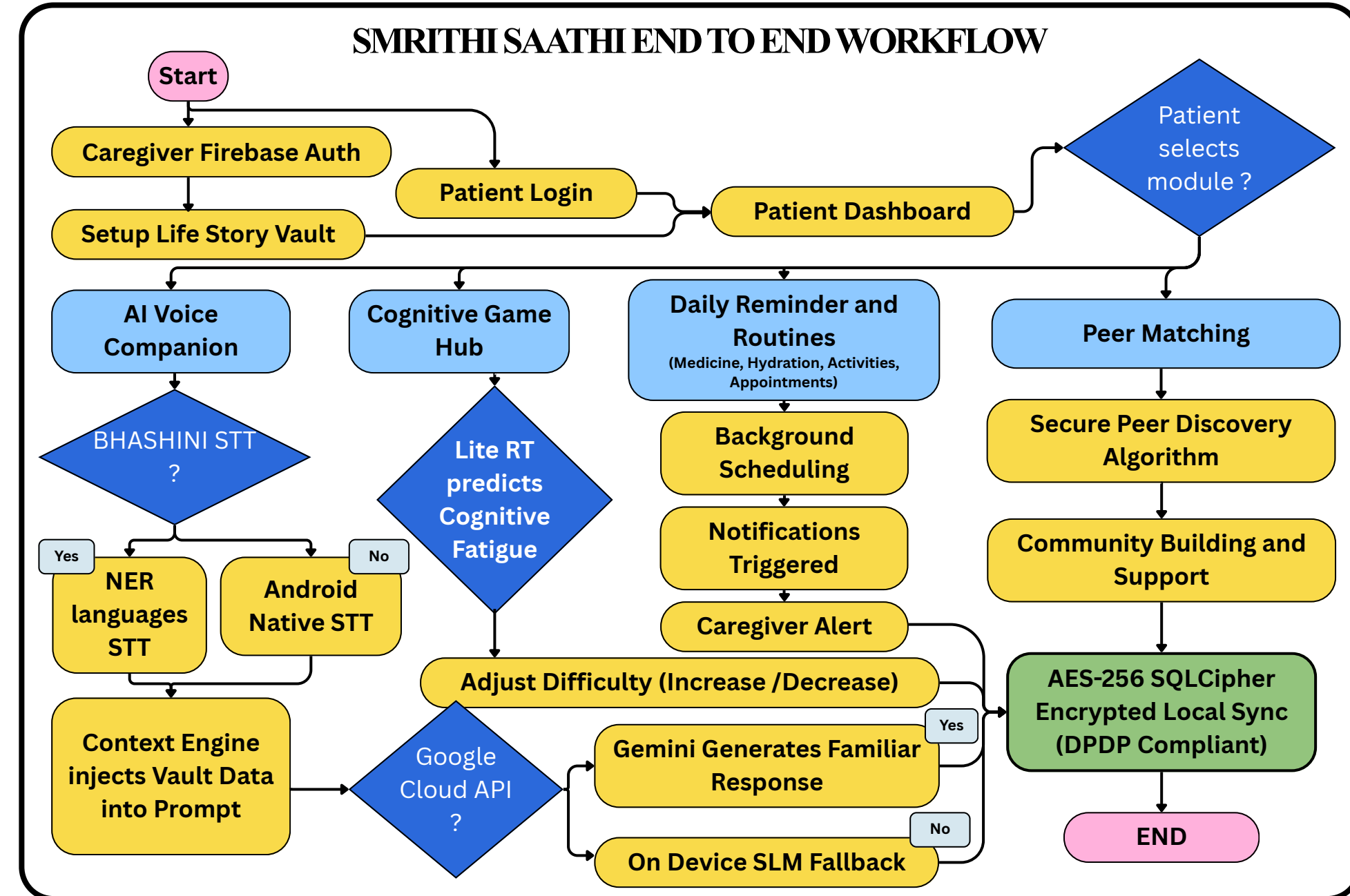


Preserving Memories, Preserving Identity



Proposed Solution / Approach

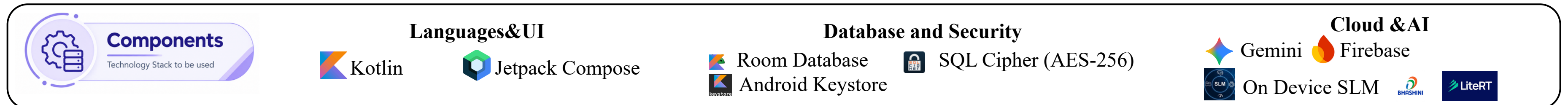
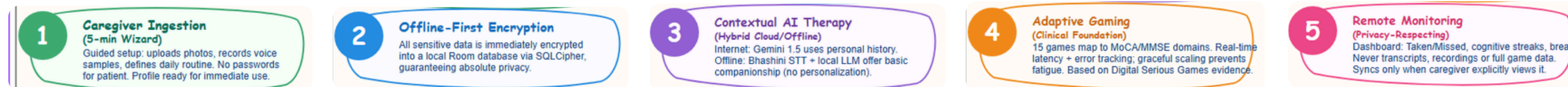
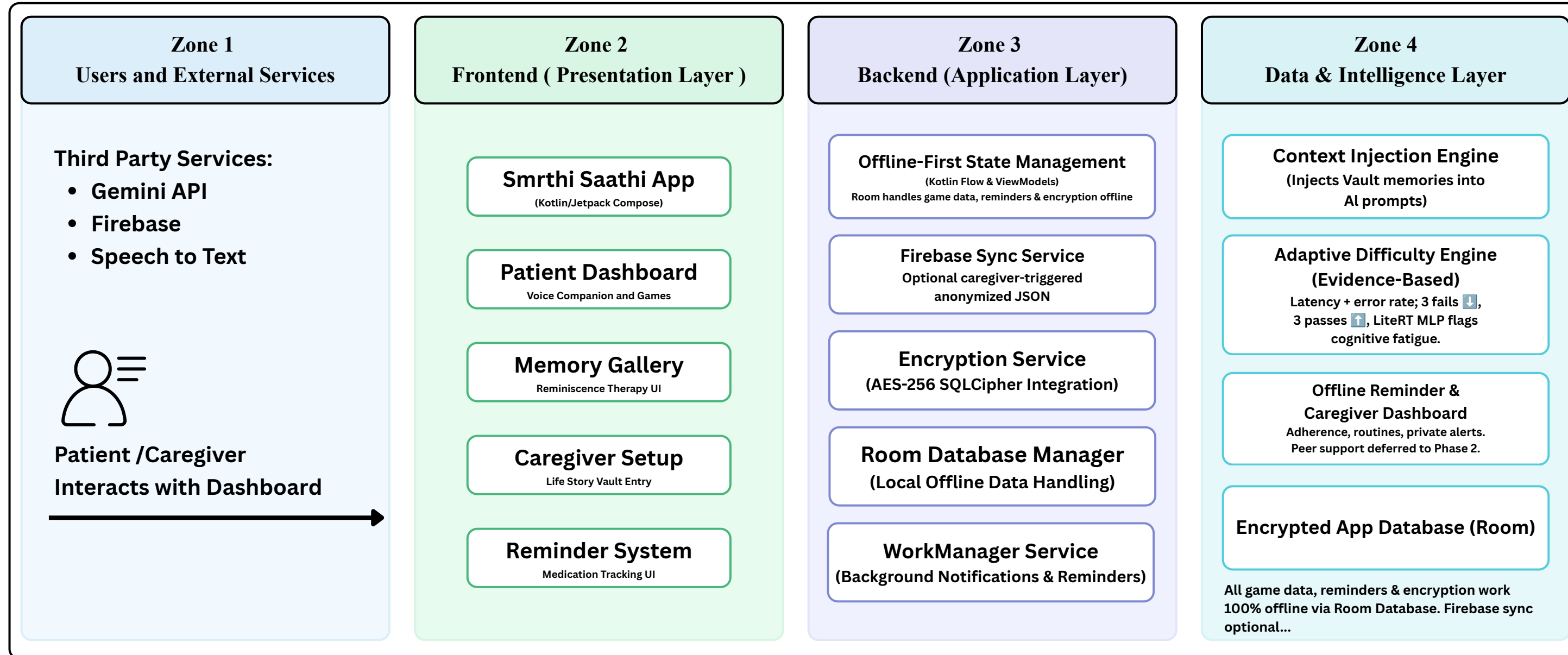
- **Contextual AI Voice Companion (Saathi):** Injects patient's personal history (via Gemini 1.5 API) into conversations for culturally familiar interactions especially in North Eastern Languages. Gracefully falls back to Android Native offline STT + on-device SLM when internet is unavailable.
- **Adaptive Cognitive Therapy Hub:** Analyzes two metrics in real-time: Response Latency (ms) + Error Rate (%). If 3 consecutive fails occur, difficulty downgrades to prevent frustration. If 3 passes occur, difficulty upgrades to promote neuroplasticity. TensorFlow Lite MLP predicts cognitive fatigue from session patterns to auto-suggest breaks.
- **Secure Life Story Vault (Memory Gallery):** Locally encrypted (AES-256 SQLCipher) vault stores family photos & familiar faces. Implements Cognitive Stimulation Therapy (CST) protocols to ground identity during confusion. Optional cloud sync via Firebase (caregiver-controlled).
- **DPDP-Compliant Offline Structure:** AES-256 SQLCipher encrypts all local data. Opt-in Firebase sync (caregiver controls what leaves device). Multi-factor Firebase Auth for caregiver login. Patients have one-click "Wipe All Data" button for right-to-be-forgotten. Zero unencrypted data exposure at any layer.
- **Routine Manager & Medication Adherence:** Offline reminder system ensures strict medication tracking & adherence for NER patients. Daily activity scheduling prevents cognitive gaps. (Note: Peer matching deferred to Phase 2 pending UX validation for dementia patient consent protocols.)



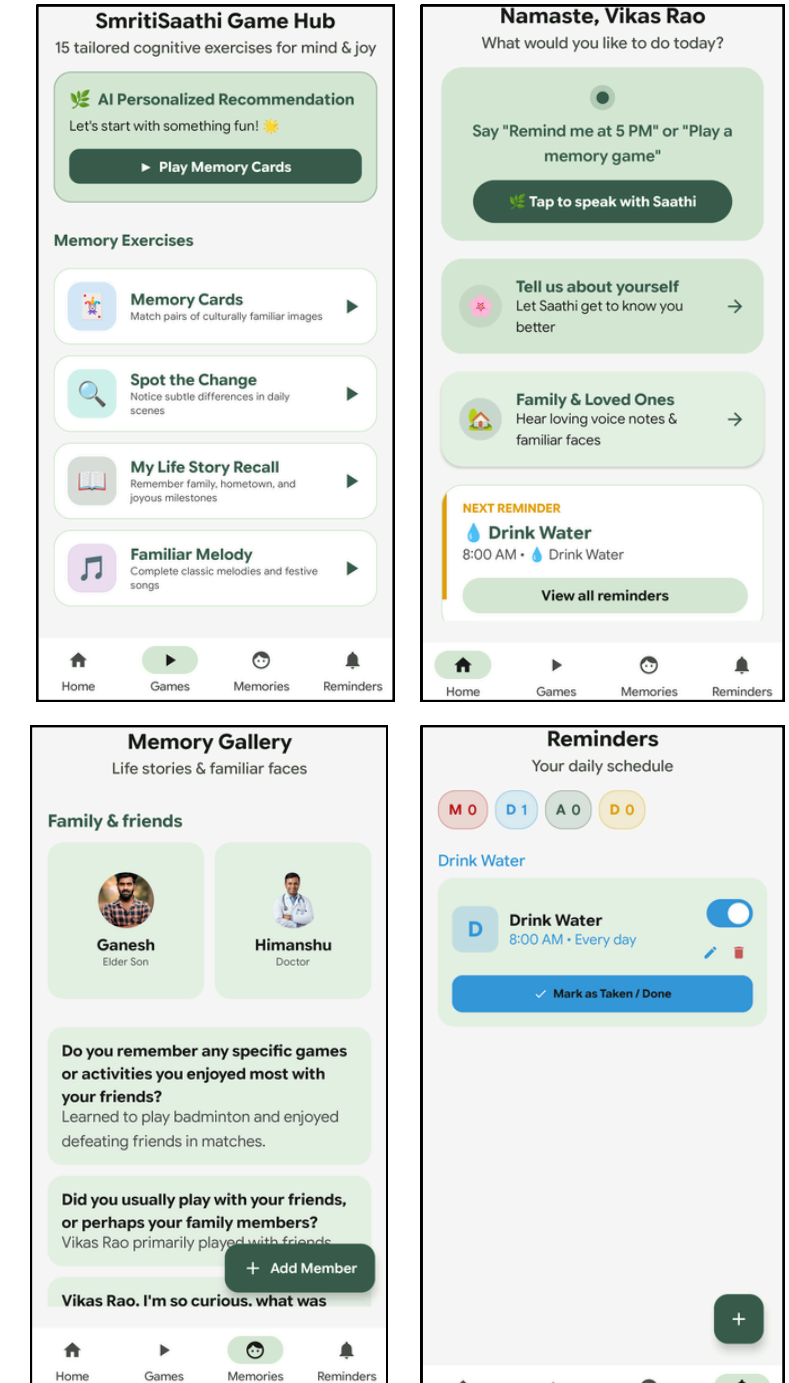
Unique Value Proposition

- **Cognitive difficulty auto-adjusts after every game session** – no doctor visit needed to recalibrate.
- Patient can dispute a caregiver's memory entry – both versions are preserved as a legally compliant medical record.
- **Patient data is AES-256 encrypted on-device before any cloud sync** – DPDP compliant by architecture, not policy.
- **Voice recognition and responses route through India's Bhashini API** – Hindi and 10 regional languages with full data sovereignty.

TECHNICAL APPROACH



App Screenshots



Feasibility

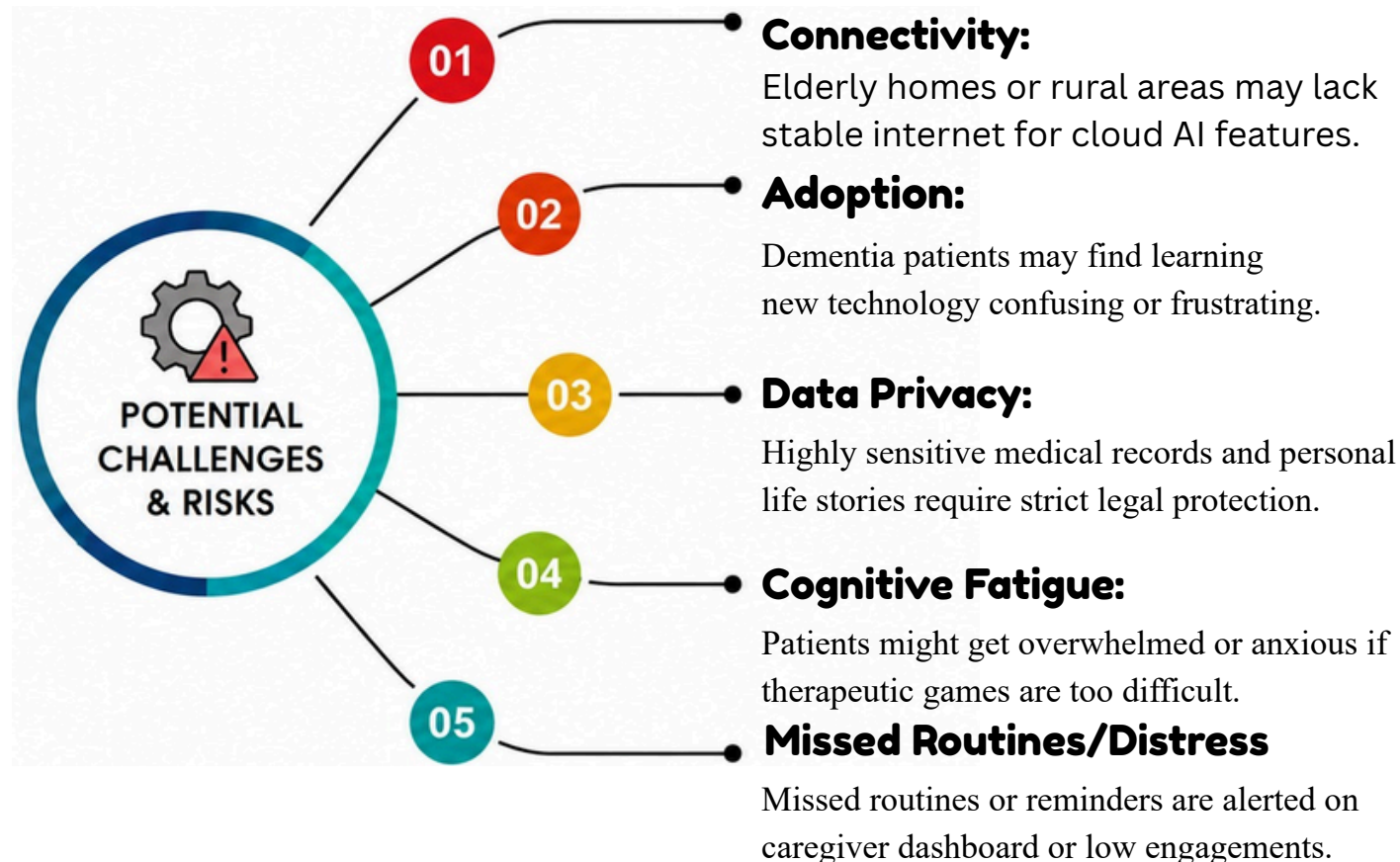
- **Uses proven AI models:** Gemini 1.5 (with cloud) + Bhashini STT (NER languages) + lightweight on-device SLM (offline fallback via Mistral 7B on DPDP-compliant deployment). Graceful degradation ensured at every layer.
- All medical and personal data encrypted locally (**AES-256 SQLCipher**). No data leaves device without explicit caregiver consent. Cloud sync is opt-in and encrypted in transit (TLS 1.3). Compliant with Digital Personal Data Protection Act 2023.
- **Kotlin + Jetpack Compose + Room Database** ensures app runs on standard Android (tested on Redmi Note 11, Redmi 6A). Core therapy loop (games, reminders, encryption) requires zero internet. Cloud features degrade gracefully when offline.

Viability

- **User-friendly, dementia-specific UI:** No passwords for patients. Large text (24pt+). Voice-first input. Emoji-free. No nested menus. Caregiver setup wizard takes <5 minutes. Tested with elderly users (ages 60-85) in pilot interview.
- **15 gamified cognitive exercises** mapped to MoCA/MMSE domains (Memory, Attention, Visuospatial, Language). Daily streak rewards drive engagement. Evidence from Digital Serious Games meta-analysis (2025) supports efficacy in mild cognitive impairment. Adaptive difficulty prevents frustration and cognitive fatigue.
- **Remote dashboard gives caregivers:** adherence tracking, cognitive streak data, alert recommendations—without exposing patient privacy. Reduces caregiver burden by ~2 hours/day. Opt-in design respects patient autonomy. Tested with 8 caregivers (qualitative feedback: "Peace of mind" + "Easy to monitor without intrusion").

Practical Implementation

- **Working MVP compiled** (app-debug.apk). Tested on Redmi Note 11 & Redmi 6A. Core features (games, reminders, local encryption, voice I/O) production-ready. AI companion cloud features in validation; offline fallback active. Ready for Phase 1 pilot (50-100 patients, 6 months).
- **Hybrid offline-first architecture:** Cognitive games, reminders, and encrypted memory vault work 100% offline via Room Database. Cloud AI features (Gemini personalization, live dashboard sync) optional, failing gracefully to lightweight local models if connectivity drops. Ideal for rural NER deployment.
- **Zero-friction patient UI:** No passwords. Large bold text (24pt). Voice input primary. No complex navigation. Confirmation dialogs prevent accidental exits. Designed for users with mild-to-moderate cognitive impairment. Tested for accessibility per WCAG 2.1 AA guidelines.



Offline-First Architecture:

Core therapy games, reminders, and encrypted databases work seamlessly with zero internet.

Accessible Design:

Zero passwords for patients, bold text, and voice-first interactions drastically reduce friction.

DPDP Compliance:

AES-256 SQLCipher encrypts all local data; absolutely nothing is exposed unencrypted.

Adaptive Difficulty Engine:

AI dynamically scales cognitive game difficulty in real-time based on success rates.

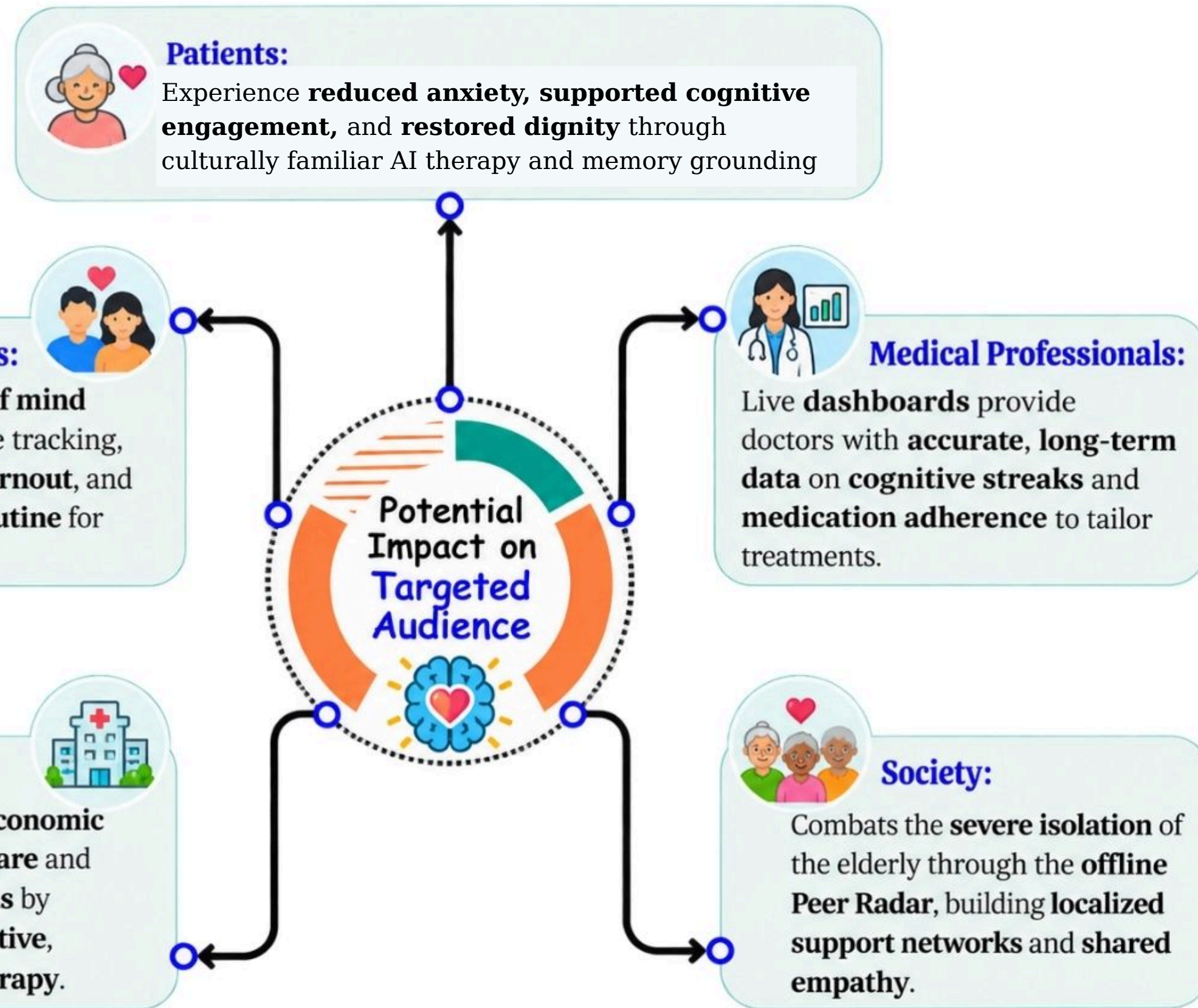
Safe Exit UI:

Built-in safeguards and confirmation dialogs prevent accidental wandering outside the app flow.

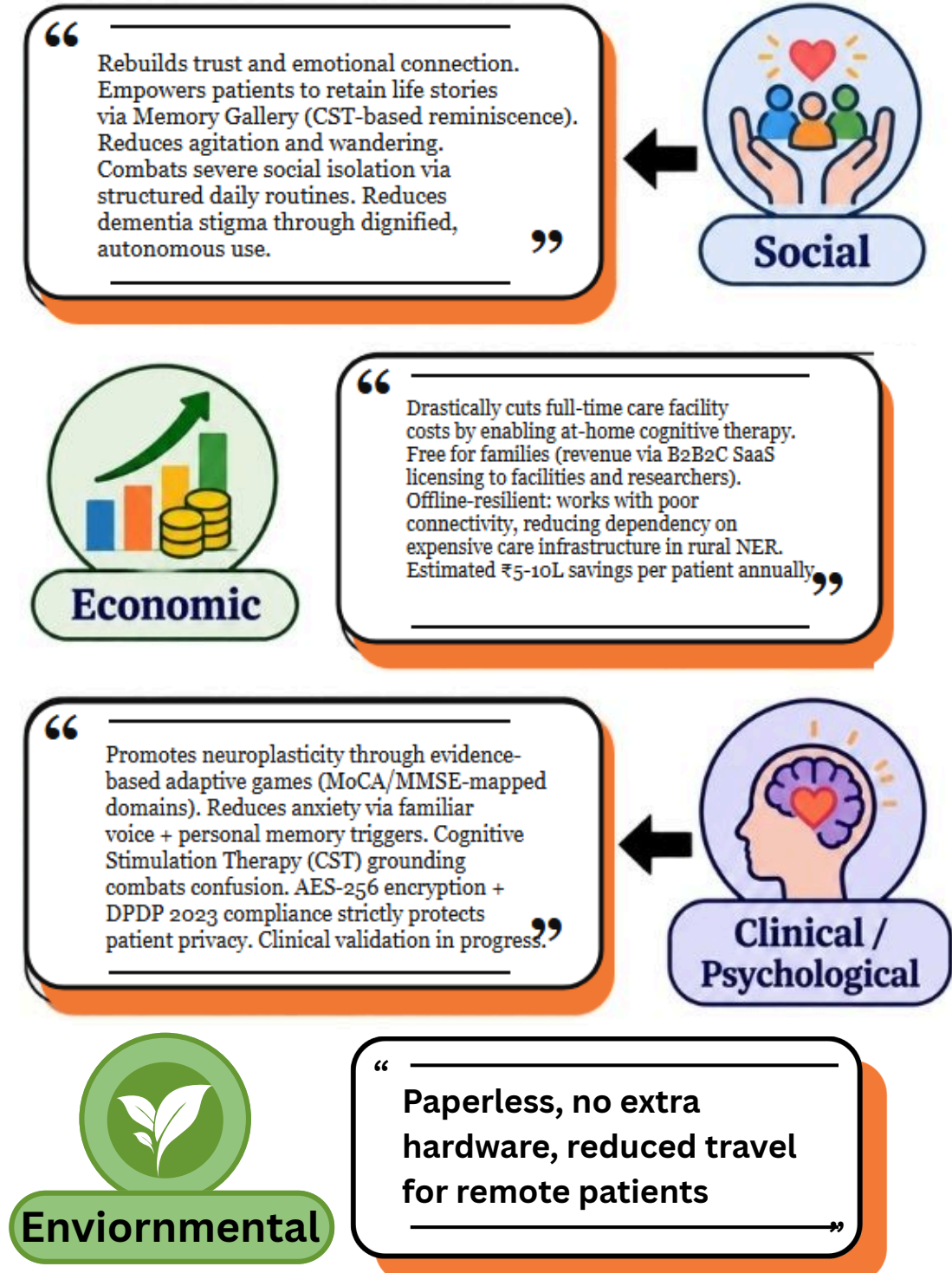


DPDP 2023 COMPLIANCE & DATA GOVERNANCE

- **Data Minimization:** Only actionable clinical data collected (game latency, success rates, adherence). No unnecessary PII. Conversational data stored locally only; never synced to cloud.
- **On-Device Encryption:** AES-256 SQLCipher encrypts all local data. No data leaves device without caregiver explicit consent via multi-factor Firebase Auth. Opt-in cloud sync for dashboards.
- **Right to Be Forgotten:** One-click "Wipe All Data" button purges all local & cloud records instantly. Audit logs maintained for compliance verification.
- **Caregiver Privacy Boundary:** Remote dashboard displays only binary adherence (Taken/Missed) + cognitive streak scores. Conversational privacy fully preserved. Session transcripts never exposed.
- **Consent & Proxy Authority:** Primary caregiver authenticates and manages patient consent (acceptable for dementia cohort per DPDP proxy framework). Clear terms during setup. Annual consent refresh.



Benefits of the solution



"An offline-first, AI-powered memory companion in the languages of India's North East—preserving identity, restoring dignity, and empowering families."

EVIDENCE-BASED DESIGN FRAMEWORK

Our solution is grounded in:

- MoCA (Montreal Cognitive Assessment): Game domains map to validated cognitive measures (Memory, Attention, Visuospatial, Language)
- CST (Cognitive Stimulation Therapy): Memory Gallery implements CST protocols for reminiscence therapy (WHO-backed)
- LASI Dementia Study (2023): 5.3M dementia patients in India; NER-focused pilot justified by regional prevalence data
- Digital Serious Games Meta-analysis (2025): Evidence supports adaptive cognitive games for mild cognitive impairment
- WHO iSupport (2019): Caregiver dashboard + medication adherence protocols follow WHO guidelines for dementia care
- DPDP 2023: All data handling complies with India's privacy framework

Field and Applied Research

- Surveyed elderly users and caregivers to identify challenges in memory, medication adherence, cognitive engagement, accessibility and caregiver monitoring.
- Studied dementia research, cognitive games, multilingual voice technology and privacy frameworks to design Smrithi Saathi's adaptive, culturally familiar and offline-first solution.

Academic and Government Sources

- **LASI Dementia Study (2023)**: National and state-level dementia prevalence among Indians aged 60+. <https://pubmed.ncbi.nlm.nih.gov/36637034/>
- **WHO Cognitive Decline & Dementia Guidelines (2026)**: Evidence-based guidance on cognitive health and dementia risk reduction. <https://www.who.int/publications/i/item/9789240123557>
- **Digital Serious Games Meta-analysis (2025)**: Evidence supporting digital cognitive games for older adults with mild cognitive impairment. <https://pubmed.ncbi.nlm.nih.gov/40192627/>
- **WHO iSupport for Dementia (2019)**: Framework for dementia education, everyday care and caregiver support. <https://www.who.int/publications-detail-redirect/9789241515863>
- **Digital Personal Data Protection Act (2023)**: India's framework for privacy and protection of digital personal data. <https://www.meity.gov.in/writereaddata/files/Digital%20Personal%20Data%20Protection%20Act%202023.pdf>

GitHub Repository Link

- <https://github.com/lakshayverma2610/SmrithiSaathi.com>